ACADEMIC PROJECT

Digital Image Processing Project

A comprehensive exploration of fundamental image transformation techniques, demonstrating how mathematical operations can reveal hidden details, enhance contrast, and transform visual information in grayscale images.

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The Original Image: A Perfect Test Case

Why This Image?

This 1280×860 grayscale seascape captured at dusk provides ideal characteristics for demonstrating various image processing techniques:

- ***Low-contrast with mid-low tones**: perfect for contrast stretching*
- ***Dark silhouettes**: ideal for negative inversion*
- ***Clear objects vs. background**: suits thresholding*
- ***Shadow details**: excellent for log transformation*
- ***Smooth gradients**: fit for gamma and piecewise adjustment*

TECHNIQUES 1–3

Fundamental Transformation Methods

1. Contrast Stretching

Maps original intensity range [min, max] → [0, 255] using linear transformation: s = 255 \cdot (r - \min) / (\max - \min). Expands the narrow intensity range to reveal hidden details. The sky becomes brighter and clearer, boats appear sharper, and water ripples emerge from mid-tones.

2. Thresholding

Applies binary transformation: g(x,y) = 1 if f(x,y) > 127, else g(x,y) = 0. Creates clean, high-contrast silhouettes by converting boats, person, and railing to solid black whilst sky and water turn pure white. Ideal for object segmentation and edge detection.

3. Image Negative

Inverts every pixel using s = 255 - r. Dark boats and person turn bright white, dim sky becomes dark grey/black, creating a surreal photographic film negative effect. Reveals details in dark-dominant regions, commonly used in X-ray analysis and medical imaging.

TECHNIQUES 4–5

Advanced Intensity Transformations

4. Log Transformation

Applies $s = c \cdot \log(1 + r)$ to expand low intensities whilst compressing high ones. Dark boats and water become significantly brighter, revealing the person's facial outline and water texture details. Sky slightly darkens to reduce glare.

> ***Purpose:*** *Expands low-intensity range, enhances shadow details — ideal for underexposed, medical, or satellite images.*

5. Power-Law (Gamma) Transformation

Applies $s = c \cdot r^{\gamma}$ with $\gamma = 0.5$ to brighten the entire image. Transforms the dusk scene into bright daylight, revealing strong water reflections and wave patterns whilst the sky retains its gradient without clipping.

> ***Purpose:*** *Corrects nonlinear human vision; used in display correction, photography, and medical imaging.*

ADVANCED TECHNIQUE

Piecewise Linear Transformation: Selective Enhancement

The Best of Both Worlds

This technique divides intensity into three segments, applying different linear stretches to each region:

01

Dark Areas [0, 70]

Stretched to [0, 0] — boats and person slightly brightened for better visibility

02

Mid-Tones [70, 140]

Stretched to [0, 180] — water and railing gain maximum contrast enhancement

03

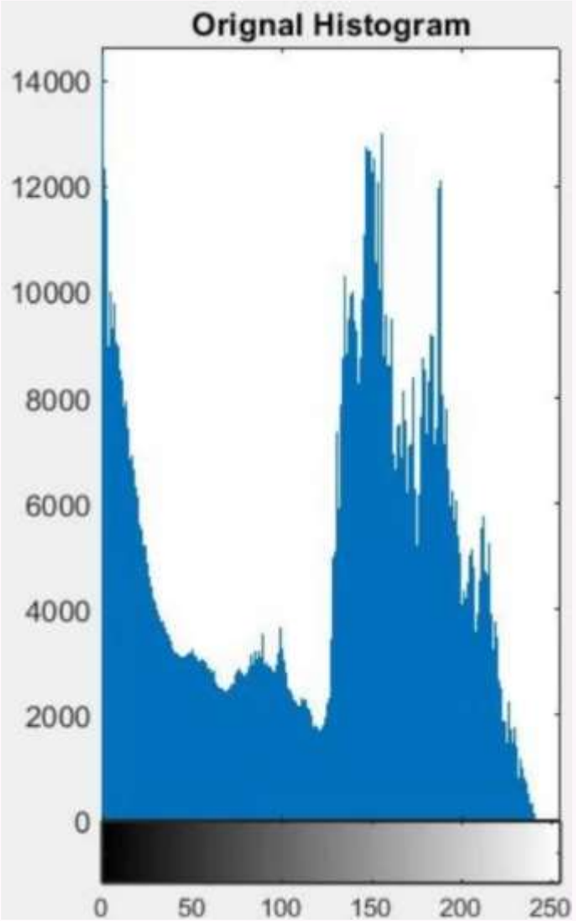
Bright Areas [140, 255]

Compressed to [180, 255] — sky preserved without overexposure

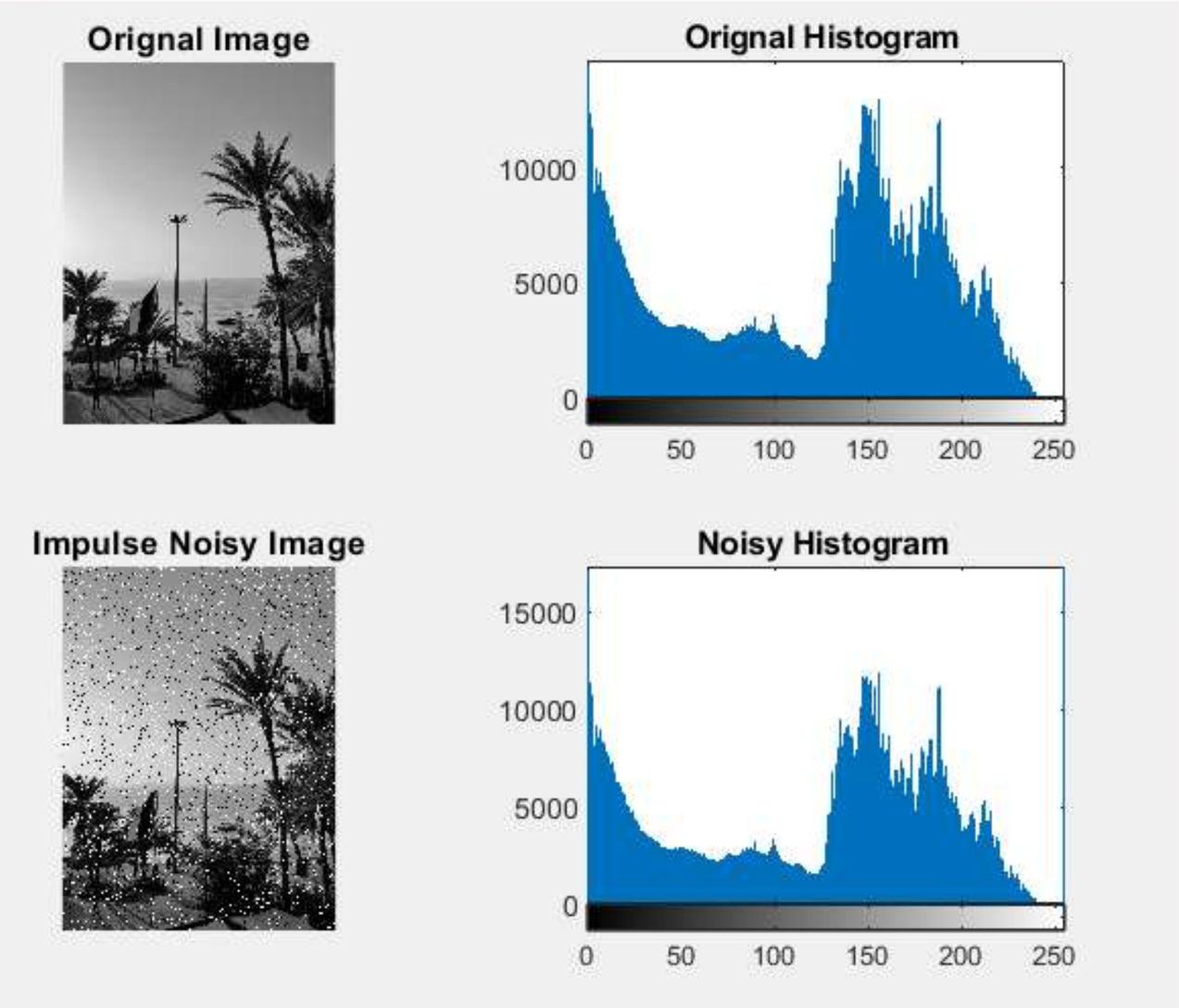
Visual Effect: *Detailed shadows with controlled highlights — water ripples become highly detailed due to mid-tone boost, whilst overall balance is maintained. Ideal for complex lighting scenes.*

Image Selection and Noise Simulation

Beach scene featuring palm trees, sand, and ocean under clear sky. Selected for diverse spatial frequencies: high-frequency details in palm fronds and sand textures, low-frequency regions in sky and water.



Histogram peaks around middle grey levels (100-150), indicating balanced distribution of light and dark areas. Critical for evaluating restoration filters.



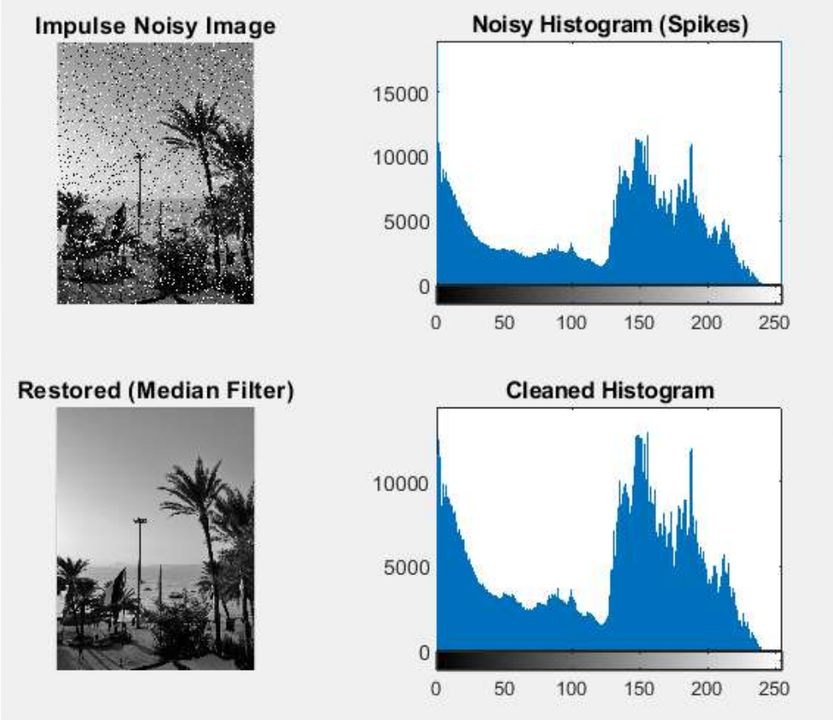
Salt & Pepper Noise

Salt & Pepper noise (5% density) was added to simulate extreme transmission errors. The histogram shows distinct spikes at 0 and 255, indicating that the image data has been replaced by pure black and white pixels, creating high-contrast corruption that obscures fine textures.

Impulse Noise Restoration Results

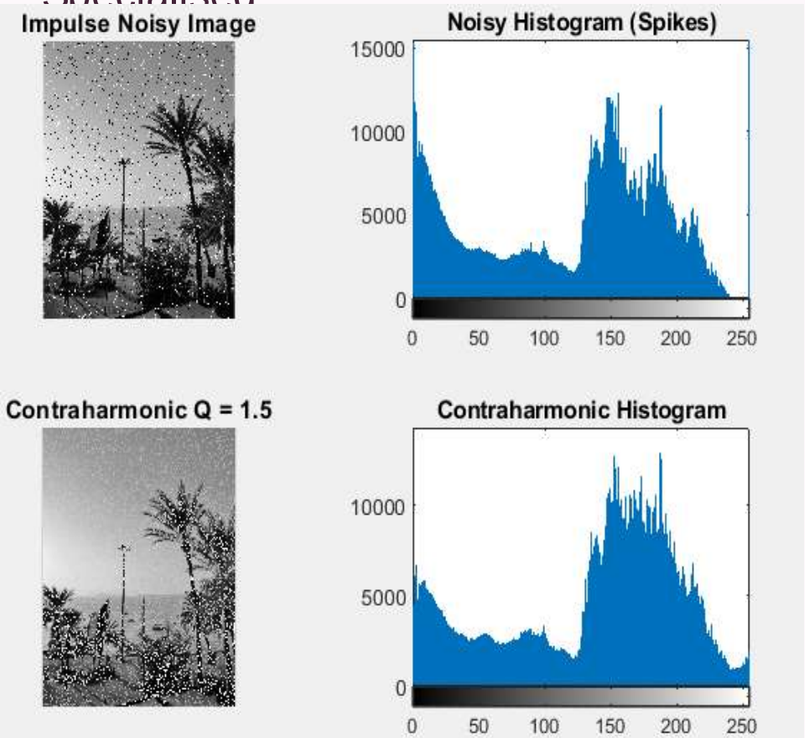
Four spatial domain filters were evaluated for removing salt and pepper noise. The median filter emerged as the gold standard, whilst linear filters proved inadequate for this noise type.

Median Filter: The Winner



Perfect restoration. Removes 100% of noise whilst preserving sharp edges in palm fronds and flags. Histogram shows complete elimination of 0/255 spikes.

Contraharmonic (Q=1.5): Specialised



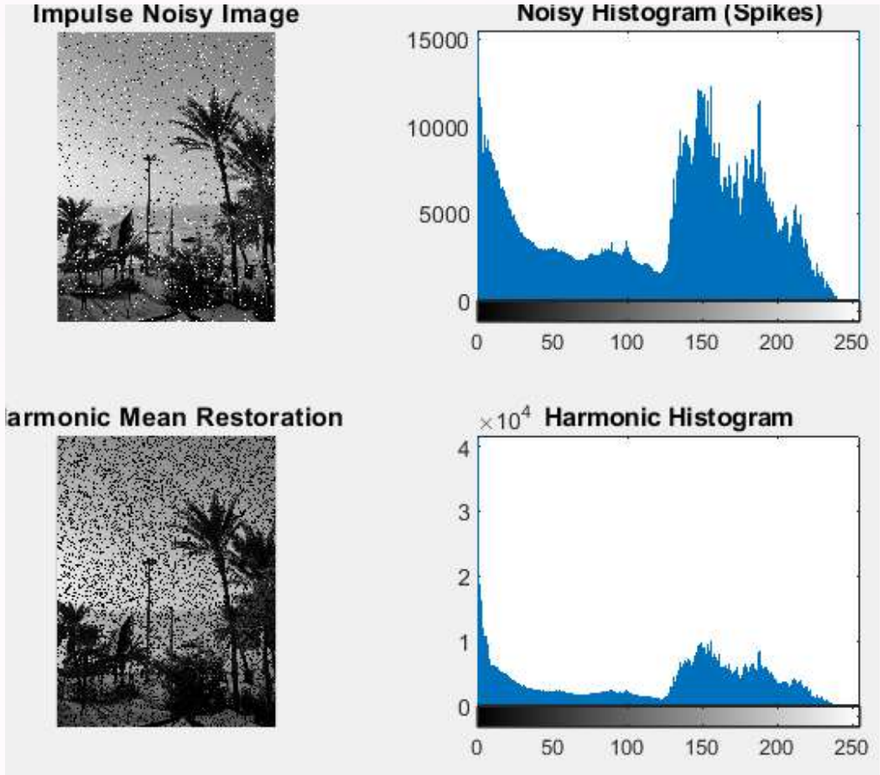
Effectively removes pepper (black dots) but leaves salt (white dots). Less universal than median filter.

Statistical Justification: Median filter maintained mean of 119.15, demonstrating noise removal without brightness shift. Histogram verification confirms spikes completely removed, returning to original smooth curves.

Impulse Noise Restoration Results

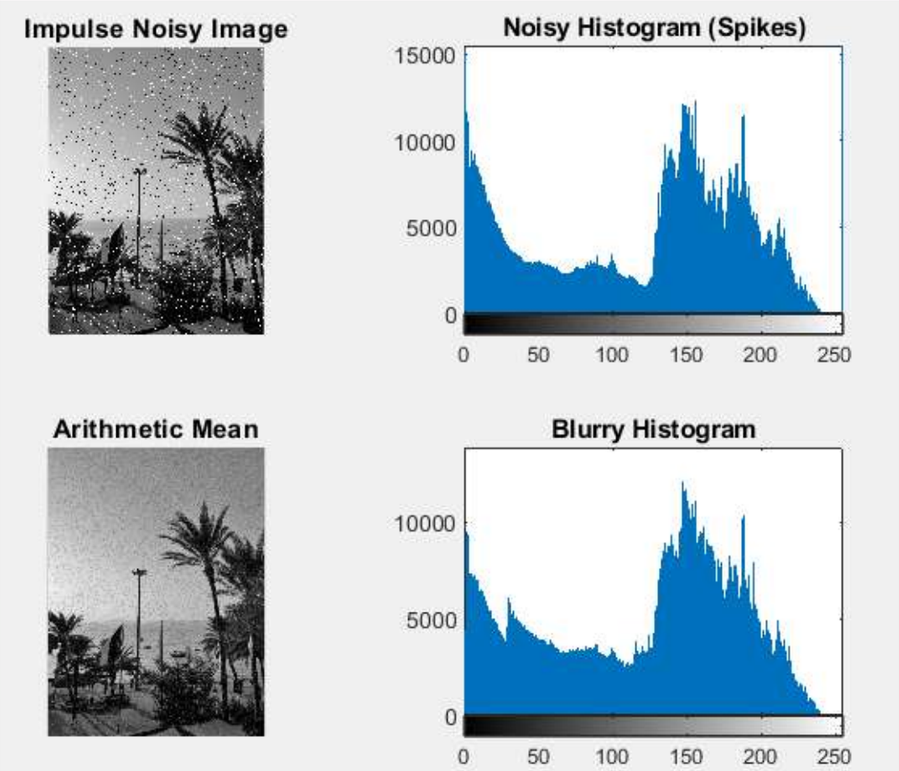
Four spatial domain filters were evaluated for removing salt and pepper noise. The median filter emerged as the gold standard, whilst linear filters proved inadequate for this noise type.

Harmonic Mean: Failure



Division-by-zero mathematics causes extreme darkening. Unsuitable for pepper noise containing 0-value pixels.

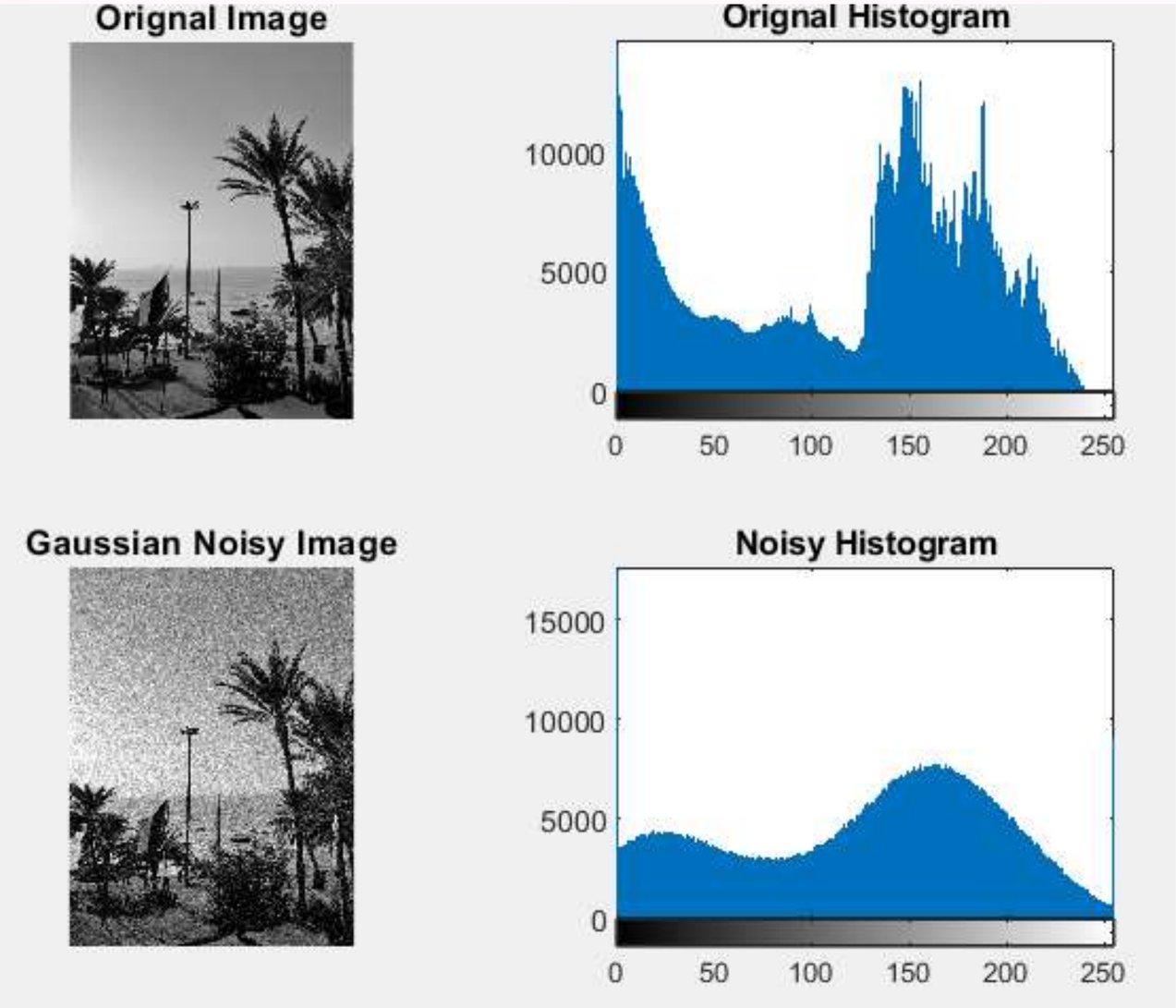
Arithmetic Mean: Failure



Smears noise into grey clouds, destroying palm and mountain textures. Linear filters wrong tool for impulse noise.

Statistical Justification: Median filter maintained mean of 119.15, demonstrating noise removal without brightness shift. Histogram verification confirms spikes completely removed, returning to original smooth curves.

Image Selection and Noise Simulation



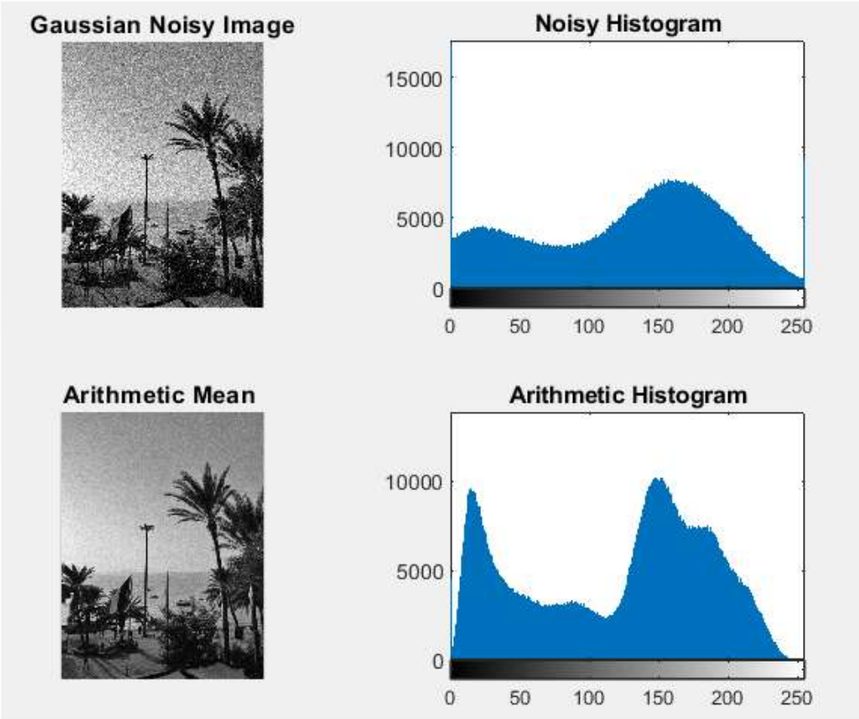
Gaussian Noise

Mean 0, variance 0.01 added to simulate electronic circuit interference. Creates grainy texture with wider, fuzzy histogram peaks.

Gaussian Noise Restoration Results

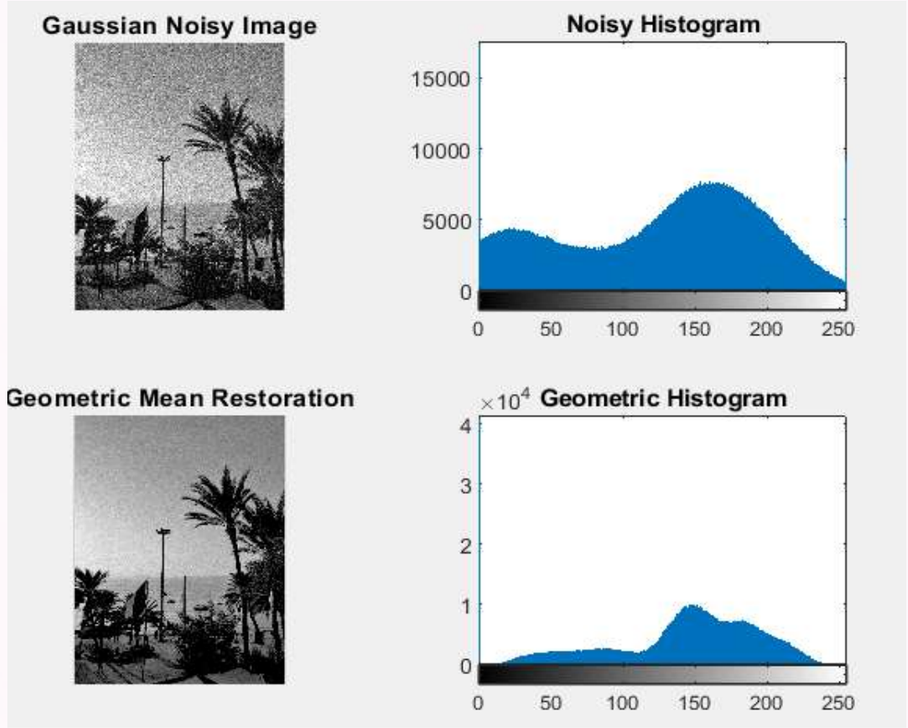
Four filters evaluated for random grain removal. Arithmetic mean filter achieved optimal smoothing through variance cancellation, whilst non-linear filters showed varying trade-offs between smoothing and detail preservation.

Arithmetic Mean: The Winner



Best for grain. Achieved lowest standard deviation (65.15). Random fluctuations cancel through averaging, producing smoothest histogram and sky appearance.

Geometric Mean: Good

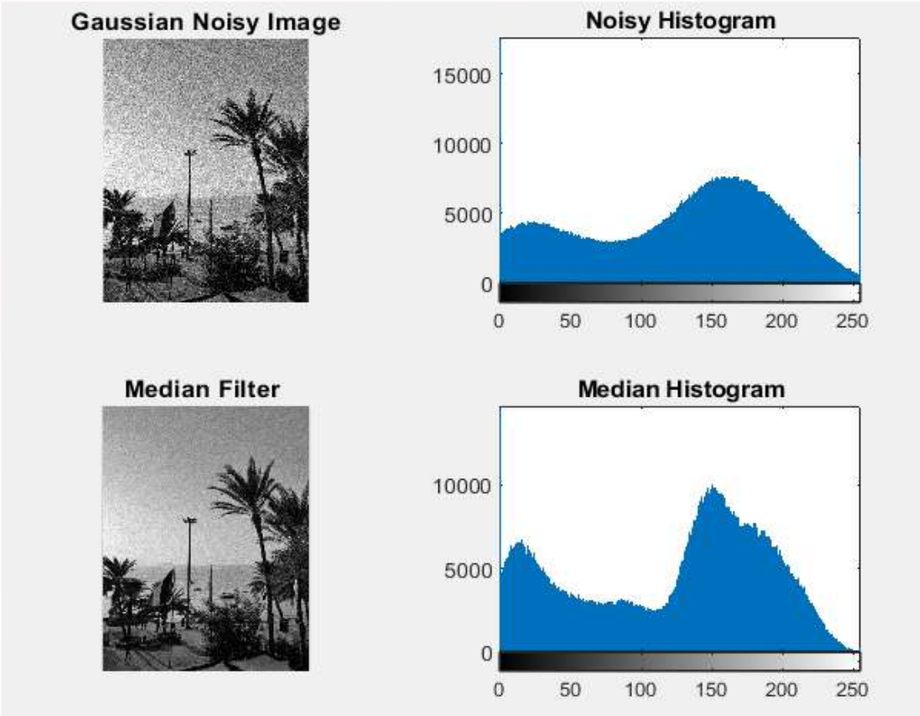


Sharper edges than arithmetic mean but residual noise in darker palm tree areas. Better detail preservation with moderate smoothing.

Gaussian Noise Restoration Results

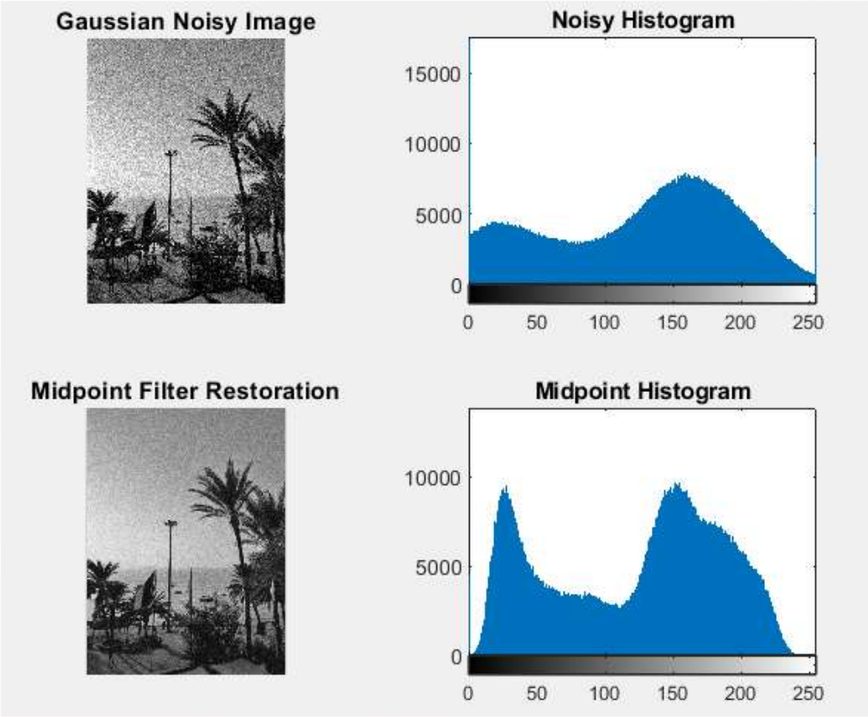
Four filters evaluated for random grain removal. Arithmetic mean filter achieved optimal smoothing through variance cancellation, whilst non-linear filters showed varying trade-offs between smoothing and detail preservation.

Median Filter: Fair



Preserves flag and palm frond edges well but creates patchy, watercolour texture in smooth sky regions. Ranking-based approach less suited to Gaussian noise.

Midpoint Filter: Moderate



High variance due to ignoring 7 of 9 pixels per window. Less effective smoothing than arithmetic mean.

> ***Final Verdict:*** *For sharp outliers, median filter is best as it ignores extreme values. For random grain, arithmetic mean filter is best as averaging cancels fluctuations. The beak/palm test proves median keeps details crisp whilst arithmetic introduces slight blur.*

Acne Removal Through Digital Image Processing

PHASE 3 RESEARCH PROJECT

DIGITAL IMAGE PROCESSING

ABSTRACT

Solving the Texture Loss Problem

Digital skin retouching often suffers from "texture loss," where removing imperfections like acne results in an unnaturally smooth surface. This research proposes a methodology to decouple skin colour from skin texture.

*By utilising a **Frequency Separation algorithm**, we transplant healthy skin "pores" onto a colour-corrected lesion site. The system is validated through real-time RGB Histogram Analysis, ensuring that the restoration remains biologically and statistically plausible.*

THEORETICAL FOUNDATION

Mapping Theory to Practice

This project bridges theoretical concepts from Digital Image Processing course slides to practical implementation, demonstrating how academic principles solve real-world challenges.

01

Digital Image Fundamentals (Ch 2)

Treating the image as a 2-dimensional signal F(x, y), converted to double precision for normalised mathematical operations.

02

Intensity Transformations (Ch 3)

Using histograms as statistical validators to ensure repaired skin matches the global distribution of the original face.

03

Spatial Domain Filtering (Ch 3)

Utilising Gaussian kernels for smoothing and "feathering" to make edits invisible through linear spatial filters.

04

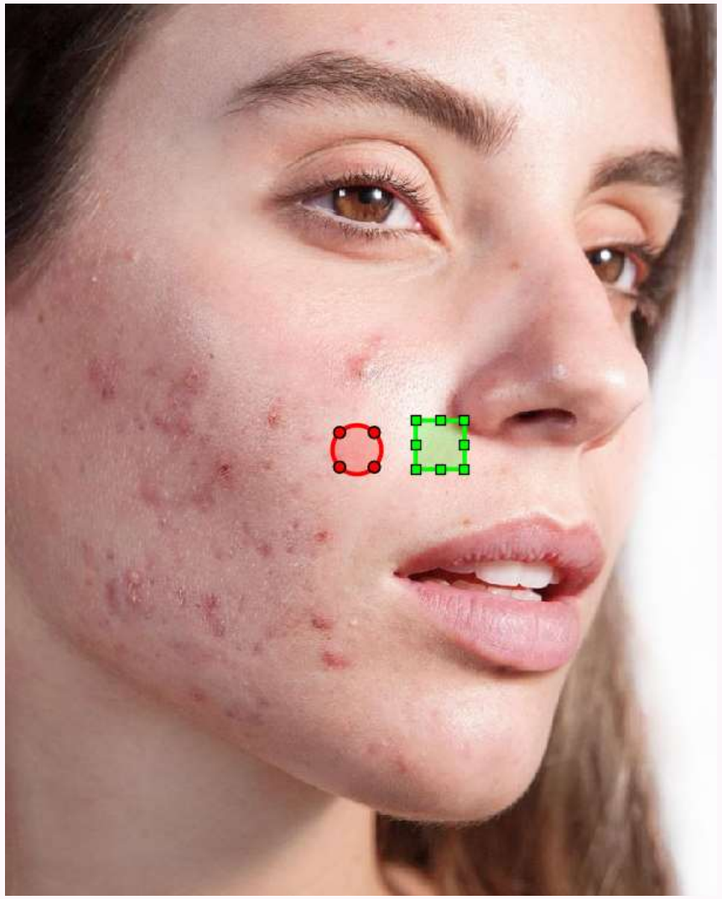
Frequency Domain Processing (Ch 4)

High-Pass Filters isolate fine details of healthy skin, whilst Low-Pass Filters remove the sharp redness of acne.

SYSTEM INTERFACE

Interactive Dermatological Restoration Environment

*The primary system interface demonstrates the **Spatial Domain Workspace** (left) which segments the "Target Set" (Red ROI) containing degradation and the "Source Set" (Green ROI) containing healthy texture. The **Spectral Fidelity Panel** (right) provides intensity histogram feedback.*



CORE ALGORITHM

The Frequency Separation Pipeline

Visual decomposition of skin into high and low-frequency layers represents the internal "Heal" engine. The image is split into distinct processing layers for optimal restoration.

Original Image

Input facial image with acne degradation

Low-Pass Layer

Colour and illumination information extracted

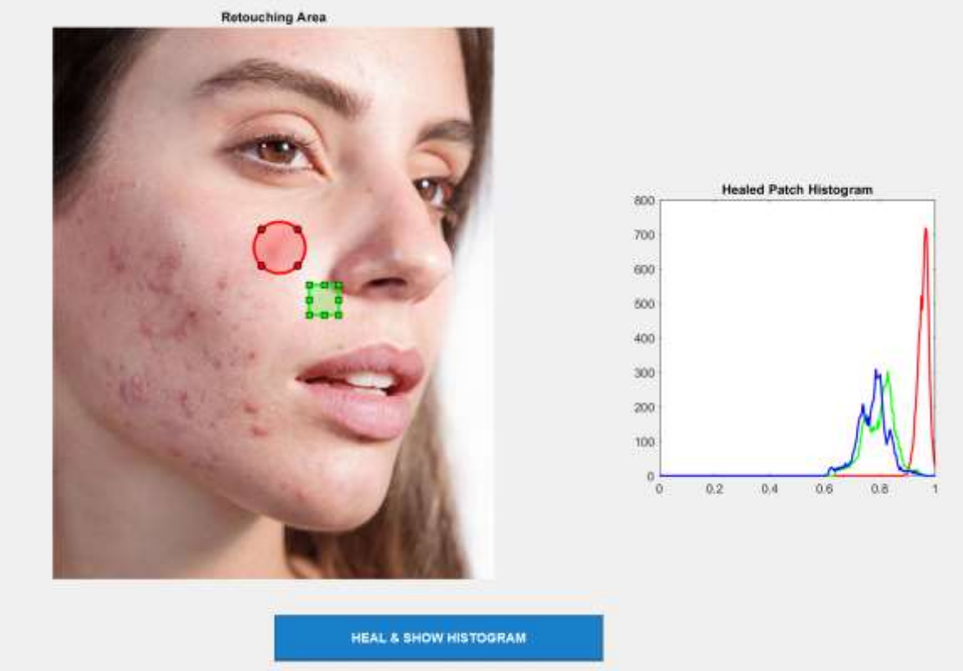
High-Pass Layer

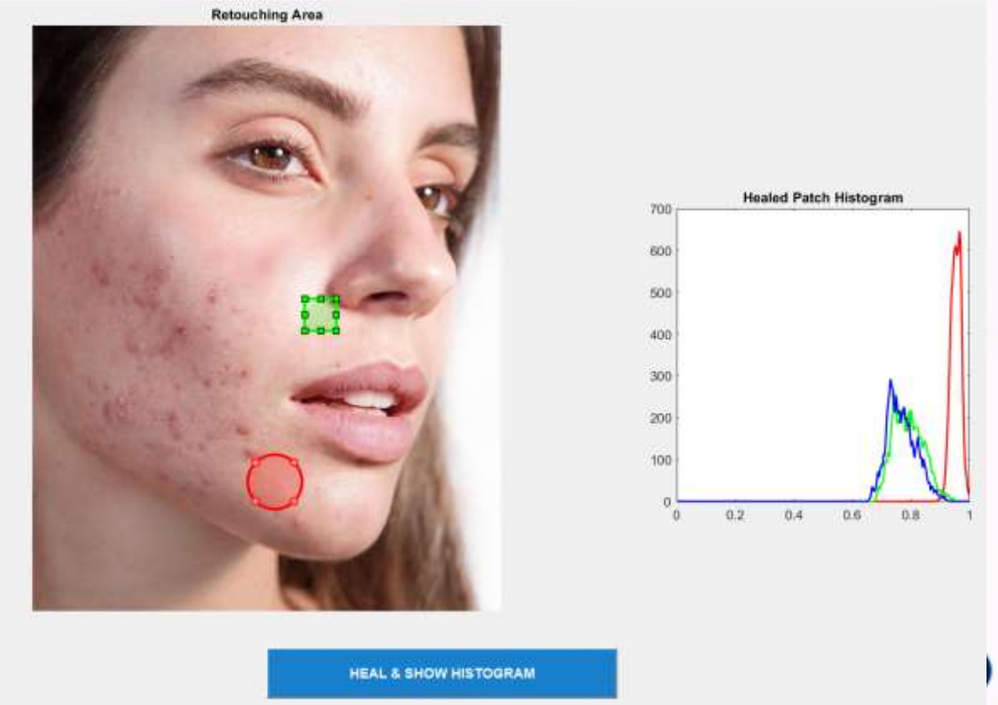
Pores and fine details isolated

Restored Output

Healthy texture replaces target colour

DIP Theory Link: *Chapter 4; HPF is used for detail extraction and LPF for noise removal.*





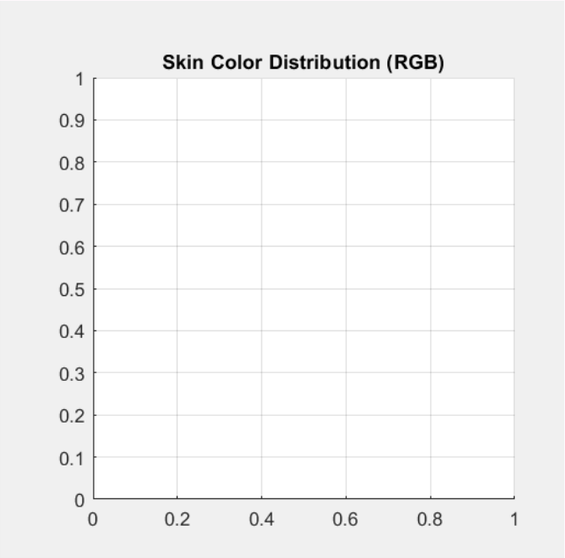
Statistical Verification Through RGB Histogram Analysis

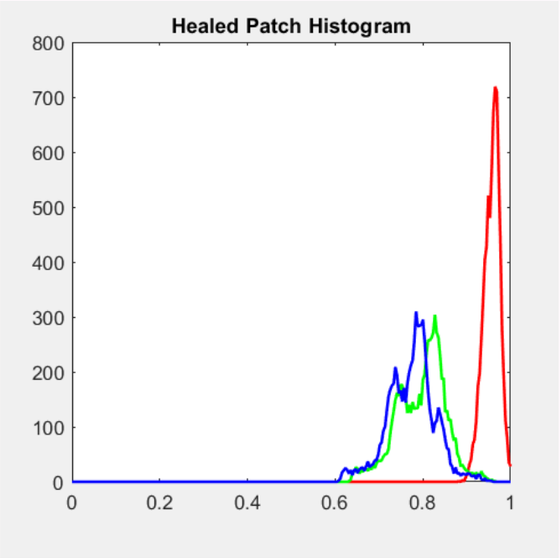
Post-Restoration Validation

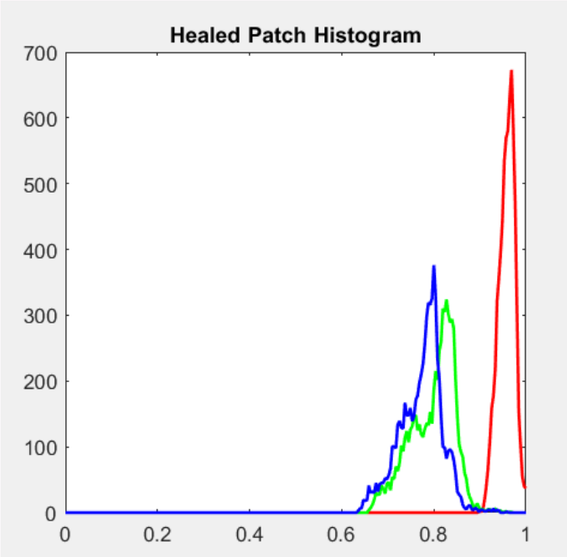
Output of the imhist function for the restored area demonstrates biological plausibility. Overlapping curves represent probability density functions for RGB channels.

A smooth histogram proves the restoration is seamless and matches the biological colour signature.

DIP Theory Link: *Chapter 3 (Spatial Domain Part 1).*







Qualitative Evaluation of the Inverse Restoration Process

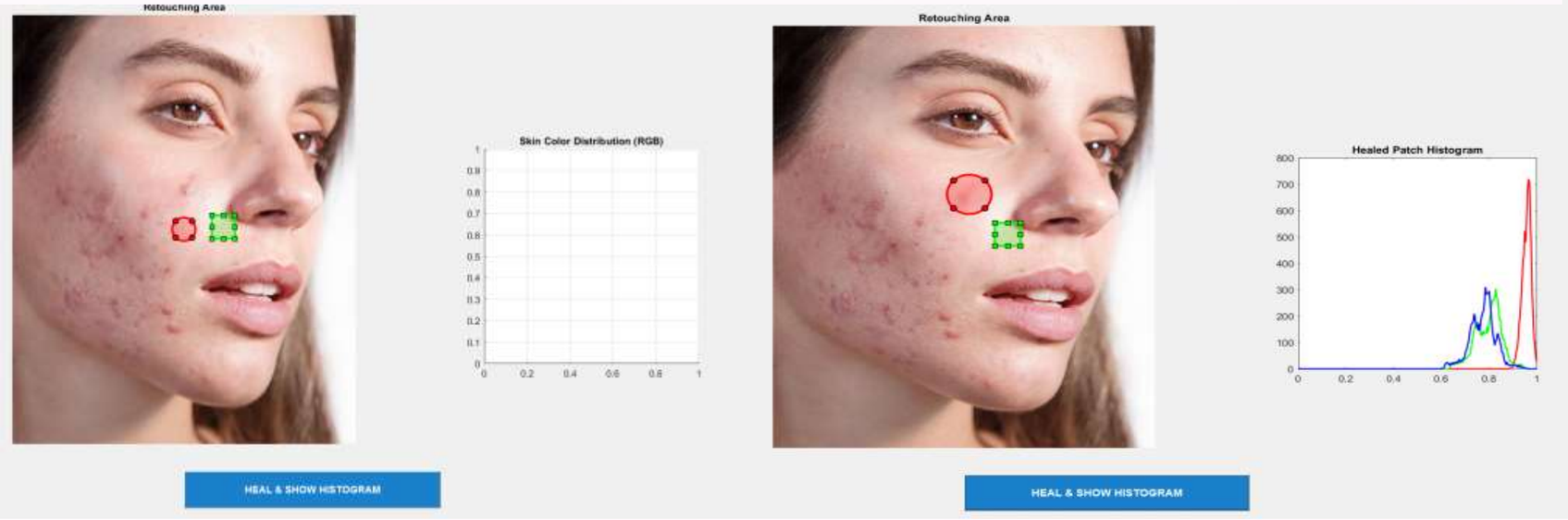
Before: Degraded State

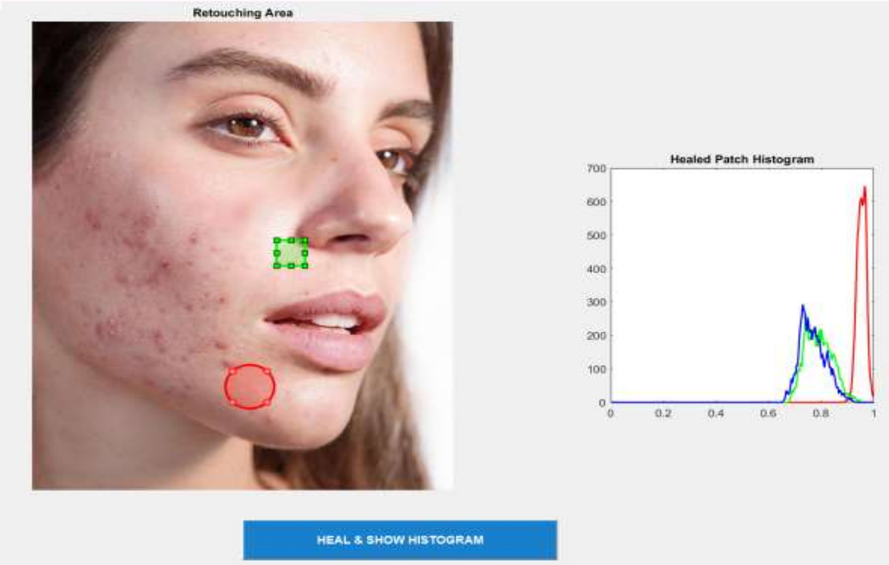
Original image showing acne degradation and texture irregularities requiring restoration.

After: Restored State

Restored image demonstrating efficacy of the Image Restoration Model with preserved biological texture.

> RESULTS


Retouching Area
Skin Color Distribution (RGB)
Retouching Area
Healed Patch Histogram
HEAL & SHOW HISTOGRAM
HEAL & SHOW HISTOGRAM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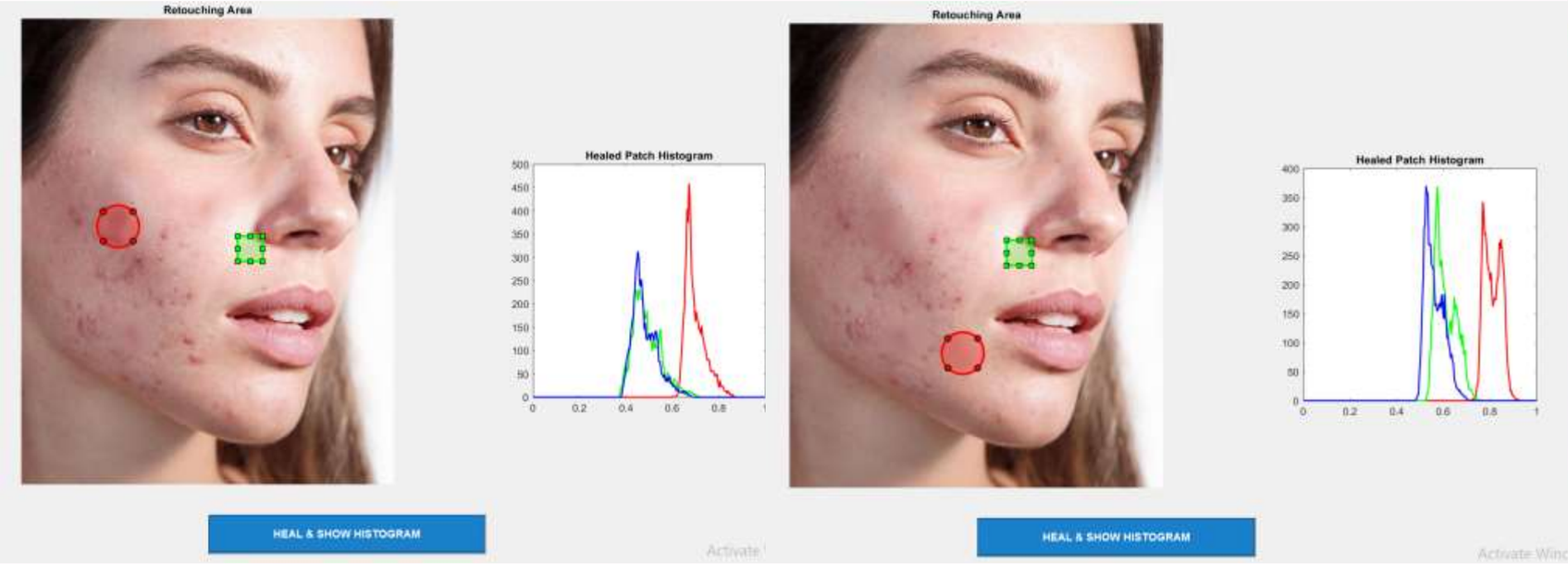
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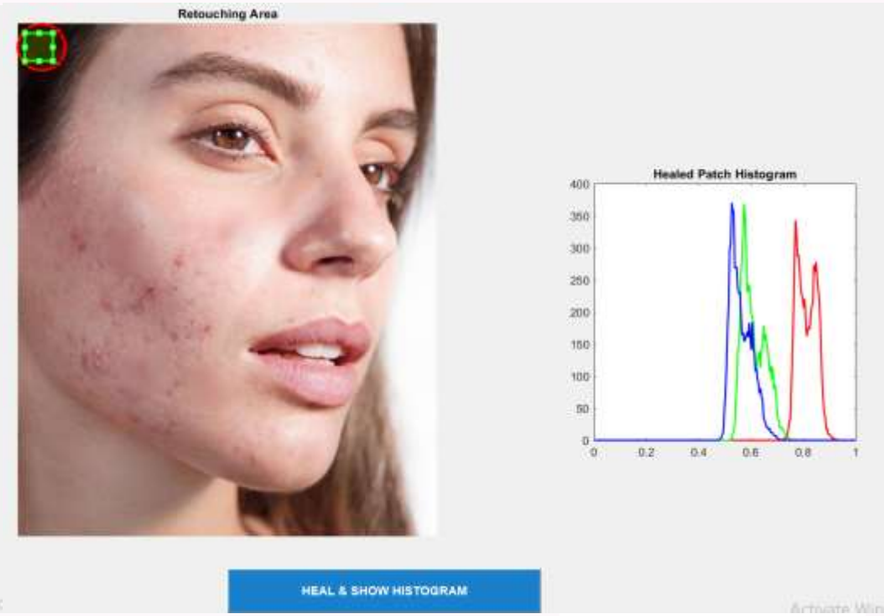
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